

Equations of First Order and First Degree

Exact Differential Equations and Reduction to Exact Equations

3.1. Exact Differential Equations. [Bombay 61; Karnatak 60]

Study the following two differential equations :

1. $x dy + y dx = 0$. Solution is $xy = C$.

2. $\sin x \cos y dy + \cos x \sin y dx = 0$.

Solution is $\sin x \cdot \sin y = C$.

We see that these differential equations can be obtained by directly differentiating their solutions. Differential equations of this type are called exact equations and bear the following property :

An exact differential equation can always be obtained from its primitive directly by differentiation, without any subsequent multiplication, elimination etc.

3.2. Necessary and Sufficient Condition

To find the necessary and sufficient condition for a differential equation of first degree being exact.

[Poona 63, 61; Delhi Hons. 57, 55; Nag. 63;
Gujrat 59; Bombay 61]

Let the equation be $M + N \frac{dy}{dx} = 0$.

... (1)

Let $u = C$ be its primitive.

... (2)

If (1) is exact, it can be obtained by directly differentiating its primitive.

Differentiating (2), we have $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dx} = 0$.

... (3)

Comparing (1) and (3) we get $M = \frac{\partial u}{\partial x}$ and $N = \frac{\partial u}{\partial y}$, so that

$$\frac{\partial M}{\partial y} = \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial N}{\partial x} = \frac{\partial^2 u}{\partial x \partial y}.$$

Hence the condition is $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

That the condition is necessary has been proved. Now we prove that it is sufficient also, i.e. if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then we show that

$M + N \frac{dy}{dx} = 0$ or $M dx + N dy = 0$ is an exact equation.

Let $\int M dx = U$, then $\frac{\partial U}{\partial x} = M$, so that

$$\frac{\partial^2 U}{\partial y \partial x} = \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \text{ as } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x},$$

$$\text{i.e. } \frac{\partial N}{\partial x} = \frac{\partial}{\partial x} \left(\frac{\partial U}{\partial y} \right)$$

Integrating, $N = \frac{\partial U}{\partial y} + f(y)$, where $f(y)$ is a function of y free from x .

$$\begin{aligned} \therefore M + N \frac{dy}{dx} &= \frac{\partial U}{\partial x} + \left[\frac{\partial U}{\partial y} + f(y) \right] \frac{dy}{dx} \\ &= \frac{d}{dx} \left[U + \int f(y) \frac{dy}{dx} dx \right] \\ &= \frac{d}{dx} [U + F(y)]. \end{aligned}$$

This shows that $M + N \frac{dy}{dx} = 0$ is an exact equation.

3.3. Working Rule (Remember it).

✓ If the equation $M dx + N dy = 0$ satisfies the condition

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x},$$

then it is exact. To integrate it,

- (i) integrate M with regard to x regarding y as constant;
- (ii) find out those terms in N which are free from x and integrate them with regard to y ;
- (iii) add the two expressions so obtained and equate the sum to an arbitrary constant.

This gives the general solution of the given exact equation.

Ex. 60. $(y^3 + 4x^3y + 3x) dx + (x^3 + 4xy^3 + y + 1) dy = 0$ [Karnatak 60]

Solution Here $M = y^3 + 4x^3y + 3x$ and $N = x^3 + 4xy^3 + y + 1$.

$$\frac{\partial M}{\partial y} = 4y^3 + 4x^3 \text{ and } \frac{\partial N}{\partial x} = 4x^3 + 4y^3.$$

Since these are equal, the equation is exact.

To find solution of the differential equation, integrating M i.e. $y^3 + 4x^3y + 3x$ w.r.t. x , keeping y as constant, we get $y^3x + x^4y + \frac{3}{2}x^2$.

In $x^4 + 4xy^3 + y + 1$, terms free from x are $y + 1$ whose integral with respect to y is $\frac{1}{2}y^2 + y$.

Therefore the general solution is

$$y^4x + x^4y + \frac{2}{3}x^2 + \frac{1}{2}y^2 + y = C.$$

Ex. 2. Solve $x(x^2 + y^2 - a^2) dx + y(x^2 - y^2 - b^2) dy = 0$.
[Nag. 63; Poona 61]

Solution. Here $M = x^3 + xy^2 - a^2x$, $N = yx^2 - y^3 - b^2y$.

$$\frac{\partial M}{\partial y} = 2xy \text{ and } \frac{\partial N}{\partial x} = 2xy.$$

Since these are equal, the equation is exact,

Integrating M w.r.t. x keeping y as constant, we get

$$\frac{1}{4}x^4 + \frac{1}{2}x^2y^2 - \frac{1}{2}a^2x^2.$$

In N , terms free from x are $-y^3 - b^2y$ whose integral is $-\frac{1}{4}y^4 - \frac{1}{2}b^2y^2$.

Hence the general solution is

$$\frac{1}{4}x^4 + \frac{1}{2}x^2y^2 - \frac{1}{2}a^2x^2 - \frac{1}{4}y^4 - \frac{1}{2}b^2y^2 = \text{const.}$$

or $x^4 - y^4 + 2x^2y^2 - 2a^2x^2 - 2b^2y^2 = C.$

Ex. 3. Solve $(x^2 - 2xy + 3y^2) dx + (4y^3 + 6xy - x^2) dy = 0$.
[Delhi Hons. 55]

Solution. Here $\frac{\partial M}{\partial y} = -2x + 6y$, $\frac{\partial N}{\partial x} = 6y - 2x$.

Since these are equal the equation is exact.

Integrating M , i.e. $x^2 - 2xy + 3y^2$ w.r.t. x keeping y as constant, we get

$$\frac{1}{3}x^3 - x^2y + 3y^2x$$

In N , term free from x is $+4y^3$ whose integral is y^4 .

Hence the solution is $\frac{1}{3}x^3 - x^2y + 3y^2x + y^4 = C$.

Ex. 4. Solve $(x - 2e^y) dy + (y + x \sin x) dx = 0$. [Gujrat 61]

Solution. Here $M = y + x \sin x$, $N = x - 2e^y$.

$\therefore \frac{\partial M}{\partial y} = 1$, $\frac{\partial N}{\partial x} = 1$; therefore equation is exact.

Integrating $y + x \sin x$ with respect to x keeping y as constant, we get $xy + \int x \sin x dx = xy - x \cos x + \sin x$.

In N , term free from x is $-2e^y$ whose integral with respect to y is $-2e^y$.

Hence the complete solution is

$$xy - x \cos x + \sin x - 2e^y = C.$$

*Ex. 5. (a) Solve $x dx + y dy = \frac{a^2(x dy - y dx)}{x^2 + y^2}$.

Solution. The equation can be put as

$$\left(x + \frac{a^2y}{x^2 + y^2}\right) dx + \left(y - \frac{a^2x}{x^2 + y^2}\right) dy = 0.$$

[Delhi Hons. 62]

Here $M = x + \frac{a^2 y}{x^2 + y^2}$ and $N = y - \frac{a^2 x}{x^2 + y^2}$.

$$\therefore \frac{\partial M}{\partial y} = \frac{(x^2 + y^2) a^2 - a^2 y \cdot 2y}{(x^2 + y^2)^2} = \frac{a^2 (x^2 - y^2)}{(x^2 + y^2)^2}$$

and $\frac{\partial N}{\partial x} = \frac{-a^2 (x^2 + y^2) + 2a^2 x^2}{(x^2 + y^2)^2} = \frac{a^2 (x^2 - y^2)}{(x^2 + y^2)^2}$.

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the equation is exact.

Integrating M w.r.t. x regarding y as constant, we get

$$\frac{1}{2}x^2 + a^2 y \frac{1}{y} \tan^{-1} \frac{x}{y} \text{ or } \frac{1}{2}x^2 + a^2 \tan^{-1} \frac{x}{y}.$$

In N , term free from x is y whose integral is $\frac{1}{2}y^2$.

Hence the solution is $\frac{1}{2}x^2 + a^2 \tan^{-1} \frac{x}{y} + \frac{1}{2}y^2 = \text{const.}$

or $x^2 + y^2 + 2a^2 \tan^{-1} \frac{x}{y} = C.$

Ex. 5. (b) Solve $x dx + y dy + \frac{x dy - y dx}{x^2 + y^2} = 0$.

The equation is exact; proceed as in the above example.

*Ex. 6. Solve $(1 + e^{x/y}) dx + e^{x/y} (1 - x/y) dy = 0$.

[Karnatak 61; Bombay 50; Gujrat 59; Poona 61]

Solution. Here $M = 1 + e^{x/y}$ and $N = e^{x/y} (1 - x/y)$

$$\frac{\partial M}{\partial y} = e^{x/y} \left(-\frac{x}{y^2} \right)$$

and $\frac{\partial N}{\partial x} = e^{x/y} \frac{1}{y} \left(1 - \frac{x}{y} \right) + e^{x/y} \left(-\frac{1}{y} \right) = e^{x/y} \left(-\frac{x}{y^2} \right).$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the equation is exact.

Now integrating $1 + e^{x/y}$ with respect to x keeping y as constant, we get $x + \frac{e^{x/y}}{1/y}$ i.e., $x + ye^{x/y}$

In N i.e., in $e^{x/y} (1 - x/y)$ there is no term free from x .

Hence the required solution is $x + ye^{x/y} = C$.

Ex. 7. $[\cos x \tan y + \cos (x+y)] dx + [\sin x \sec^2 y + \cos (x+y)] dy = 0$.

[Bombay 61; Gujrat 61]

Solution. Here $M = \cos x \tan y + \cos (x+y)$,

and $N = \sin x \sec^2 y + \cos (x+y)$.

Now $\frac{\partial M}{\partial y} = \cos x \sec^2 y - \sin (x+y)$,

$$\frac{\partial N}{\partial x} = \cos x \sec^2 y - \sin(x-y).$$

Since these are equal, the equation is exact.

Now integrating M , i.e. $\cos x \tan y + \cos(x+y)$ with respect to x keeping y as constant, we get

$$\sin x \tan y + \sin(x+y)$$

In N , there is no term free from x .

Hence the general solution is

$$\sin x \tan y + \sin(x+y) = C.$$

Ex. 8. $(\cos x \tan y - \sin x \sec y) dx$
 $+ (\sin x \sec^2 y + \cos x \tan^2 y \operatorname{cosec} y) dy = 0.$
 [Bombay B. A. (Sub.) 58]

Solution. We have $M = \cos x \tan y - \sin x \sec y$,
 and $N = \sin x \sec^2 y + \cos x \tan^2 y \operatorname{cosec} y$.

$$\therefore \frac{\partial M}{\partial y} = \cos x \sec^2 y - \sin x \sec y \tan y$$

$$\frac{\partial N}{\partial x} = \cos x \sec^2 y - \sin x \tan y \sec y.$$

$$\text{as } \tan^2 y \operatorname{cosec} y = \tan y \sec y.$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the equation is exact.

Integrating M i.e. $\cos x \tan y - \sin x \sec y$ with regard to x keeping y as constant we get

$$\sin x \tan y + \cos x \sec y.$$

In N there is no term free from x .

Hence the general solution is

$$\sin x \tan y + \cos x \sec y = C.$$

Ex. 9. Solve $(\sin x \cos y + e^{2x}) dx$
 $+ (\cos x \sin y + \tan y) dy = 0.$
 [Poona 59]

Solution. Here $\frac{\partial M}{\partial y} = -\sin x \sin y$, $\frac{\partial N}{\partial x} = -\sin x \sin y$.

Since these are equal, the equation is exact.

Integrating M i.e., $\sin x \cos y + e^{2x}$ w.r.t. x , keeping y as constant, we get $-\cos x \cos y + \frac{1}{2}e^{2x}$.

Also in N the term free from x is $\tan y$ whose integral w.r.t. y is $\log \sec y$.

Hence the solution is

$$-\cos x \cos y + \frac{1}{2}e^{2x} + \log \sec y = C.$$

Ex. 10. Solve the following equations (which are exact):
 $(2x^3 + 3y) dx + (3x + y - 1) dy = 0.$

Ans. $\frac{1}{2}x^4 + 3yx - \frac{1}{2}y^2 - y = C$
 [Poona 93.]

$$(ii) (x^2 - 4xy - 2y^2) dx + (y^2 - 4xy - 2x^2) dy.$$

$$\text{Ans. } x^3 + y^3 - 6xy(x+y) = C.$$

$$(iii) \cos x (\cos x - \sin a \sin y) dx$$

$$+ \cos y (\cos y - \sin a \sin x) dy = 0.$$

$$\text{Ans. } 2(x+y) \sin 2x + \sin 2y - 4 \sin a \sin x \sin y = C.$$

$$(iv) (2xy + y - \tan y) dx + (x^2 - x \tan^2 y + \sec^2 y) dy = 0.$$

[Poona 1964]

$$\text{Ans. } x^2y + xy - x \tan y + \tan y = C.$$

$$(14) (2x^2y + 4x^3 - 12xy^2 + 3y^2 - xe^y + e^{2x}) dy$$

$$+ (12x^2y + 2yx^2 + 4x^3 - 4y^3 + 2ye^{2x} - e^y) dx = 0. \quad [\text{Poona 64}]$$

$$\text{Ans. } 4x^3y + x^2y^2 + x^4 - 4y^3x + ye^{2x} - xe^y + y^3 = C.$$

3.4. Integrating factors.

If an equation becomes exact after it has been multiplied by a function of x and y , then such a function is called an integrating factor

[Karnatak 61]

3.5. Number of integrating factors.

To show that there is an infinite number of integrating factors for an equation.

$$M dx + N dy = 0.$$

[Karnatak 61]

To prove this let μ be an integrating factor: then

$$\mu (M dx + N dy) = du.$$

Integrating, $u = c$ is a solution.

Now multiplying both the sides by $f(u)$, a function of u , we get

$$\mu f(u) [M dx + N dy] = f(u) du.$$

Expression on the right is directly integrable and therefore so must be the left hand side.

Hence $\mu f(u)$ is also an integrating factor. Since $f(u)$ is an arbitrary function of u , the number of integrating factors is infinite.

3.6. Integrating factor by inspection.

Sometimes an integrating factor can be found by inspection. For this the reader should study the following results:—

Group of terms

I F.

Exact Differential

$$x dy - y dx$$

$$\frac{1}{x^2}$$

$$\frac{x dy - y dx}{x^2} = d\left(\frac{y}{x}\right)$$

$$x dy - y dx$$

$$\frac{1}{y^2}$$

$$\frac{y dx - x dy}{-y^2} = d\left(-\frac{x}{y}\right)$$

$$x dy - y dx$$

$$\frac{1}{xy}$$

$$\frac{dy}{y} - \frac{dx}{x} = d\left(\log \frac{y}{x}\right)$$

$$x dy - y dx$$

$$\frac{1}{x^2 + y^2}$$

$$\begin{aligned} \frac{x dy - y dx}{x^2 + y^2} &= \frac{\frac{x dy - y dx}{x^2}}{1 + \left(\frac{y}{x}\right)^2} \\ &= d\left[\tan^{-1} \frac{y}{x}\right] \end{aligned}$$

Groups of terms	I.F.	Exact Differential
$x dy + y dx$	$\frac{1}{(xy)^n}$	$\frac{x dy + y dx}{xy} = d [\log (xy)]$ for $n=1$
$x dx + y dy$	$\frac{1}{(x^2+y^2)^n}$	$\frac{x dx + y dy}{(x^2+y^2)^n}$ $= d \left[-\frac{1}{2(n-1)(x^2+y^2)^{n-1}} \right]$ or $= \frac{x dx + y dy}{x^2+y^2} = d \left[\frac{1}{2} \log (x^2+y^2) \right]$ if $n=1$.

Ex. 1. Solve $(x+y^2) dy + (y-x^2) dx = 0$.

[Nagpur 61]

Solution. The equation can be written as

$$x dy + y dx + y^2 dy - x^2 dx = 0.$$

$$\text{or } d(xy) + y^2 dy - x^2 dx = 0.$$

Integrating, $xy + \frac{1}{3}y^3 - \frac{2}{3}x^3 = A$ or $y^3 - x^3 + 3xy = c$.

Ex. 2. Solve $y dx - x dy + 3x^2y^2e^{x^3} dx = 0$.

[Nagpur 61]

Solution. The equation can be written as

$$\frac{y dx - x dy}{y^2} + 3x^2e^{x^3} dx = 0,$$

$$d\left(\frac{x}{y}\right) + e^{x^3} d(x^3) = 0.$$

Integrating, $\frac{x}{y} + e^{x^3} = c$.

Ex. 3. Solve $x dy - y dx - x(x^2 - y^2)^{1/2} dx = 0$. [Delhi Hons. 61]

Solution. The equation can be written as

$$\frac{x dy - y dx}{(x^2 - y^2)^{1/2}} - x dx = 0$$

$$\frac{x dy - y dx}{x^2}$$

$$\text{i.e. } \frac{\frac{x dy - y dx}{x^2}}{\left[1 - \left(\frac{y}{x}\right)^2\right]^{1/2}} = dx, \text{ put } \frac{y}{x} = t, \text{ then } \frac{x dy - y dx}{x^2} = dt$$

$$\text{i.e. } \frac{dt}{\sqrt{1-t^2}} = dx \text{ or } x + c = \sin^{-1} t = \sin^{-1} \left(\frac{y}{x}\right).$$

Ex. 4. Solve $a(x dy + 2y dx) = xy dy$.

Solution. The equation can be written as

$$(a-y)x dy + 2ay dx = 0 \text{ or } \frac{a-y}{y} dy + \frac{2a}{x} dx = 0.$$

Integrating, $a \log y - y + 2a \log x = C$.

or $\log yx^2 = \frac{y}{a} + \log C$ or $yx^2 = Ce^{y/a}$.

Ex. 5. Solve $y dx - x dy + \log x dx = 0$.

Solution. The equation is $x \frac{dy}{dx} - y = \log x$

or $\frac{dy}{dx} - \frac{1}{x} y = \frac{\log x}{x}$. Linear, I.F. = $e^{-\int \frac{1}{x} dx} = \frac{1}{x}$.

$$\therefore y \cdot \frac{1}{x} = \int \frac{1}{x^2} \log x dx - C$$

$$= -\frac{1}{x} (1 + \log x) - C$$

or $y + \log x + Cx + 1 = 0$ is the solution.

Ex. 6. Solve $(1+xy) y dx + (1-xy) x dy = 0$. [Bihar 62]

Solution. Write the equation as

$$y dx + x dy + xy (y dx - x dy) = 0$$

or $d(xy) + xy (y dx - x dy) = 0$.

We readily find that $\frac{1}{x^2 y^2}$ is the I.F. So the equation becomes

$$\frac{d(xy)}{x^2 y^2} + \frac{y dx - x dy}{xy} = 0 \text{ or } \frac{d(xy)}{(xy)^2} + \left(\frac{dx}{x} - \frac{dy}{y} \right) = 0.$$

Integrating, $-\frac{1}{xy} + \log x - \log y = C_1$ or $x = Cye^{1/xy}$.

Ex. 7. Solve $(x^4 e^x - 2mxy^2) dx + 2mx^2 y dy = 0$.

Solution. Equation is $2y \frac{dy}{dx} - \frac{2y^2}{x} + \frac{x^2 e^x}{m} = 0$.

Putting $y^2 = z$, the equation becomes $\frac{dz}{dx} - \frac{2}{x} z + \frac{x^2 e^x}{m} = 0$.

I.F. = $e^{-\int \frac{2}{x} dx} = \frac{1}{x^2}$, etc.

Ex. 8. Solve $y(2xy + e^x) dx - e^x dy = 0$.

[Vikram 61]

Solution. The equation is $e^x \frac{dy}{dx} = 2xy^2 + ye^x$

or $-y^{-2} \frac{dy}{dx} + y^{-1} = -2xe^{-x}$. Put $y^{-1} = v$, $-y^{-2} \frac{dy}{dx} = \frac{dv}{dx}$.

$\therefore$ the equation is $\frac{dv}{dx} + v = -2xe^{-x}$. I.F. = e^x etc.

Solution is $y^{-1} e^x = -x^2 + C$.

13.7. Rules for finding the integrating factor. ✓

Rule I. If $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x)$, a function of x only, then $e^{\int f(x) dx}$ is an integrating factor. [Delhi Hons. 64]

Rule II. If $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{M} = g(y)$ is a function of y alone, then $e^{\int -g(y) dy}$ is an integrating factor.

We give below some examples to illustrate these rules.

Ex. 1. Solve $(x^2 + y^2 + x) dx + xy dy = 0$.

Solution $M = x^2 + y^2 + x$, $N = xy$.

$\frac{\partial M}{\partial y} = 2y$, $\frac{\partial N}{\partial x} = y$, equation is not exact.

However, $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2y - y}{xy} = \frac{1}{x}$, a function of x alone.

Hence I.F. $= e^{\int \frac{1}{x} dx} = e^{\log x} = x$.

Multiplying by I.F., the equation becomes

$(x^3 + xy^2 + x^2) dx + x^2y dy = 0$, exact now (check up).

Integrating, $x^3 + xy^2 + x^2$ with regard to x , keeping y as constant, we get $\frac{1}{4}x^4 + \frac{1}{2}x^2y^2 + \frac{1}{3}x^3$

and in x^2y^2 there is no term free from x . Therefore the solution is $\frac{1}{4}x^4 + \frac{1}{2}x^2y^2 + \frac{1}{3}x^3 = C'$ or $3x^4 + 4x^3 + 6x^2y^2 = C$.

Ex. 2. Solve $(x^2 + y^2 + 1) dx - 2xy dy = 0$.

Solution. $\frac{\partial M}{\partial y} = 2y$, $\frac{\partial N}{\partial x} = -2y$, not exact.

However, $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2y + 2y}{-2xy} = -\frac{2}{x}$ function of x alone.

$\therefore$ I.F. $= e^{-\int \frac{2}{x} dx} = e^{-2 \log x} = \frac{2}{x^2}$.

Multiplying by $\frac{1}{x^2}$ the equation becomes

$\left(1 + \frac{y^2}{x^2} + \frac{1}{x^2}\right) dx - \frac{2y}{x} dy = 0$, exact now.

Integrating, $1 + \frac{y^2}{x^2} + \frac{1}{x^2}$ with regard to x keeping y as constant,

we get $x - \frac{y^2}{x} - \frac{1}{x}$

and in $-\frac{2y}{x}$ there is no term free from x .

Hence the solution is

$$x - \frac{y^2}{x} - \frac{1}{x} = C \text{ or } x^2 - y^2 = Cx + 1.$$

Ex. 3. Solve $(x^2 + y^2) dx - 2xy dy = 0$.

Solution. Just as in the above example, I.F. = $\frac{1}{x^2}$.

Hence multiplying by $\frac{1}{x^2}$ the equation becomes

$$\left(1 + \frac{y^2}{x^2}\right) dx - \frac{2y}{x} dy = 0, \text{ exact.}$$

$\therefore$ Solution is $x - \frac{y^2}{x} = c$ or $x^2 - y^2 = cx$.

Ex. 4. Solve $(x^2 + y^2 + 2x) dx + 2y dy = 0$.

[Vikram 1959 ; Alld. 59]

Solution. $\frac{\partial M}{\partial y} - 2y, \frac{\partial N}{\partial x} = 0$, not exact.

$$\text{However, } \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2y}{2y} = 1.$$

$$\therefore \text{I.F.} = e^x.$$

Multiplying by e^x , the equation becomes

$$e^x (x^2 + y^2 + 2x) dx + 2ye^x dy = 0, \text{ now exact.}$$

This can be written as

$$(x^2 + 2x) e^x dx + (y^2 e^x dx + e^x \cdot 2y dy) = 0$$

or $d(x^2 e^x) + d(y^2 e^x) = 0$.

$\therefore$ Integrating, $x^2 e^x + y^2 e^x = C$ or $(x^2 + y^2) e^x = C$

Aliter. The equation can also be written as

$$2y \frac{dy}{dx} + y^2 = -(x^2 + 2x).$$

Putting $y^2 = v$, $\frac{dv}{dx} + v = -(x^2 + 2x)$. Linear, I.F. = e^x etc

*Ex. 5. Solve $\left(\frac{1}{2}y + \frac{1}{2}y^2 + \frac{1}{2}x^2\right) dx + \frac{1}{2}(x + xy^2) dy = 0$,

[Delhi Hons. 1965 ; Agra M.Sc. 63 ; Banaras 56]

Solution. $\frac{\partial M}{\partial y} = 1 + y^2, \frac{\partial N}{\partial x} = \frac{1}{2}(1 + y^2)$, not exact.

$$\text{However, } \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{(1 + y^2) - \frac{1}{2}(1 + y^2)}{\frac{1}{2}x(1 + y^2)} = \frac{3}{x}.$$

a function of x alone

$$\therefore \text{I.F.} = e^{\int \frac{3}{x} dx} = e^{3 \log x} = x^3.$$

Multiplying by x^3 , the equation becomes

$$(x^3 y + \frac{1}{2} x^3 y^2 + \frac{1}{2} x^3) dx + \frac{1}{2} (x^4 + x^4 y^2) dy = 0, \text{ exact now.}$$

Integrating $x^3 y + \frac{1}{2} x^3 y^2 + \frac{1}{2} x^3$ with respect to x keeping y as constant, we get $\frac{1}{2} x^4 y + \frac{1}{12} x^4 y^2 + \frac{1}{6} x^4$.

In $\frac{1}{2} (x^4 + x^4 y^2)$ there is no term free from x .

$\therefore$ the solution is $\frac{1}{2} x^4 y + \frac{1}{12} x^4 y^2 + \frac{1}{6} x^4 = \text{constant}$

or

$$3x^4 y + y^2 x^4 + x^4 = C.$$

Ex. 6. Is the differential equation $(x^3 - 2y^2) dx + 2xy dy = 0$ exact? Solve the equation. [Cal. Hons. 1963]

Solution. The equation is not exact; however we have

$$\frac{1}{N} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right] = \frac{-4y - 2y}{2xy} = -\frac{3}{x}; \therefore \text{I.F.} = e^{\int -3 \frac{dx}{x}} = \frac{1}{x^3}.$$

Proceed as above.

$$\text{Ex. 7. } (2x^3 y^2 + 4x^2 y + 2xy^2 + xy^4 + 2y) dx + 2(y^3 - x^2 y + x) dy = 0.$$

Solution. Equation is not exact.

$$\frac{1}{N} \left\{ \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right\} = 2x. \text{ I.F.} = e^{\int 2x dx} = e^{x^2}.$$

The solution is $(2x^2 y^3 + 4xy + y^4) e^x = C$.

* Ex. 8. Solve $(y^4 + 2y) dx + (xy^3 + 2y^4 - 4x) dy = 0$.

[Cal. Hons. 1962, 61]

Solution. $\frac{\partial M}{\partial y} = 4y^3 + 2, \frac{\partial N}{\partial x} = y^3 - 4$, not exact

However, $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{M} = \frac{4y^3 + 2 - (y^3 - 4)}{y^4 + 2y} = \frac{3}{y}$ a function of y alone.

$$\therefore \text{I.F.} = e^{-\int \frac{3}{y} dy} = e^{-3 \log y} = \frac{1}{y^3}.$$

Multiplying by $1/y^3$, the equation becomes

$$\left(y + \frac{2}{y^2} \right) dx + \left(x + 2y - \frac{4x}{y^3} \right) dy = 0, \text{ exact now.}$$

Integrating $y + \frac{2}{y^2}$ w.r.t. x keeping y as constant, we have

$$yx + \frac{2}{y^2} x.$$

In $x + 2y - \frac{4x}{y^3}$, the term free from x is $2y$. So integrating $2y$ w.r.t. y , we get y^2 .

Therefore the solution is $yx + \frac{2}{y^2} x + y^2 = C$.

Ex. 9. Solve $(3x^2y^3 + 2xy) dx + (2x^3y^3 - x^2) dy = 0$.

[Cal. Hons. 54, 53]

Solution. Here $\frac{\partial M}{\partial y} = 12x^2y^2 + 2x$, $\frac{\partial N}{\partial x} = 6x^2y^3 - 2x$.

Now $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{M} = \frac{6x^2y^3 + 4x}{y(3x^2y^3 + 2x)} = \frac{2}{y}$ function of y alone.

$$\therefore \text{I. F.} = e^{\int \frac{2}{y} dy} = e^{2 \log y} = \frac{1}{y^2}$$

Multiplying by $\frac{1}{y^2}$, the equation becomes

$$\left(3x^2y^2 + \frac{2x}{y}\right) dx + \left(2x^3y - \frac{x^2}{y^2}\right) dy = 0, \text{ exact now.}$$

Integrating $3x^2y^2 + \frac{2x}{y}$ w.r.t. x keeping y as constant, we get

$$x^3y^2 + \frac{x^2}{y}$$

In $2x^3y - \frac{x^2}{y^2}$, there is no term free from x .

Hence the solution is $x^3y^2 + \frac{x^2}{y} = C$

$$x^3y^3 - x^2 = Cy.$$

* or Ex. 10. Solve $(2xy^4e^y + 2xy^3 + y) dx + (x^2y^4e^y - x^2y^2 - 3x) dy = 0$.

Solution. We have $\frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{4}{y} \therefore \text{I.F.} = \frac{1}{y^5}$

Solution is $x^2e^y + \frac{x^2}{y} \cdot \frac{x}{y^3} = C$.

* Rule III.

If $M dx + N dy = 0$ is homogeneous and $Mx + Ny \neq 0$,

then $\frac{1}{Mx + Ny}$ is an integrating factor.

* Rule IV.

[Delhi Hons. 61]

If the equation can be written in the form

$$yf(xy) dx + xg(xy) dy = 0, f(xy) \neq g(xy),$$

then $\frac{1}{xy[f(xy) - g(xy)]} = \frac{1}{Mx - Ny}$ is an integrating factor.

Ex. 11. Solve $x^2y dx - (x^3 + y^3) dy = 0$.

Solution. The equation is homogeneous and

$$M_x + N_y \neq 0.$$

Hence $\frac{1}{M_x + N_y} = \frac{1}{x^3y - (x^3 + y^3)y} = -\frac{1}{y^4}$ is integrating factor.

Multiplying by $-\frac{1}{y^4}$ the equation becomes

$$-\frac{x^2}{y^3} dx - \frac{x^3 + y^3}{y^4} dy = 0, \text{ exact now.}$$

Integrating $-\frac{x^2}{y^3}$ with respect to x treating y as constant,

$$\text{we get } -\frac{x^3}{3y^3}.$$

In $\frac{x^3 + y^3}{y^4}$ term free from x is $\frac{y^3}{y^4}$, i.e., $\frac{1}{y}$. Integrating it w.r.t. y , we get $\log y$.

Hence the solution is $-\frac{x^3}{3y^3} + \log y = \text{const.} = \log C$

$$\text{or } \log y = \log C + \frac{x^3}{3y^3} \quad \text{or } y = C e^{x^3/3y^3}.$$

Ex. 2 ✓ Solve $(x^4 + y^4) dx - xy^3 dy = 0$.

Solution. Equation is homogeneous.

$$\therefore \text{I.F.} = \frac{1}{M_x + N_y} = \frac{1}{x(x^4 + y^4) - xy^4} = \frac{1}{x^5}.$$

Multiplying by $\frac{1}{x^5}$, the equation becomes

$$\left(\frac{1}{x} + \frac{y^4}{x^5}\right) dx - \frac{y^3}{x^4} dy = 0, \text{ exact now.}$$

Integrating $\frac{1}{x} + \frac{y^4}{x^5}$ with respect to x keeping y constant, we get

$$\log x - \frac{y^4}{4x^4}.$$

Also $-\frac{y^3}{x^4}$ is not free from x .

Hence the complete solution is $\log x - \frac{y^4}{4x^4} = C''$.

$$y^4 = 4x^4 \log x + Cx^4.$$

Ex. 3 ✓ Solve $y^2 dx + (x^2 - xy - y^2) dy = 0$.

Solution. The equation is homogeneous.

$$\text{I.F.} = \frac{1}{M_x + N_y} = \frac{1}{y(x^2 - y^2)}$$

Multiplying by $\frac{1}{y(x^2 - y^2)}$, the equation becomes

$$\frac{y}{x^2-y^2} dx + \left(\frac{1}{y} - \frac{x}{x^2-y^2} \right) dy = 0, \text{ exact now.}$$

Integrating $\frac{y}{x^2-y^2}$ with regard to x keeping y as constant, we have $\int \frac{y}{x^2-y^2} dx = \frac{1}{2} \int \left(\frac{1}{x-y} - \frac{1}{x+y} \right) dx = \frac{1}{2} \log \frac{x-y}{x+y}$.

Also in $\frac{1}{y} - \frac{x}{x^2-y^2}$, term free from x is $\frac{1}{y}$, its integral being $\log y$. Hence the solution is

$$\frac{1}{2} \log \frac{x-y}{x+y} + \log y = \log C_1 \text{ or } (x-y)y^2 = C(x+y).$$

Ex. 63. Solve $y(xy + 2x^2y^2) dx + x(xy - x^2y^2) dy = 0$.

[Vikram 63; Calcutta Hons. 62]

Solution. Equation is of the form

$$yf(xy)dx + xg(xy)dy = 0.$$

$$\text{I.F.} = \frac{1}{Mx - Ny} = \frac{1}{xy(xy + 2x^2y^2) - xy(xy - x^2y^2)} = \frac{1}{3x^3y^3}.$$

Multiplying by $\frac{1}{3x^3y^3}$, the equation becomes

$$\left(\frac{1}{3x^2y} + \frac{2}{3x} \right) dx + \left(\frac{1}{3xy^2} - \frac{1}{3y} \right) dy = 0 \text{ (exact now).}$$

Integrating $\frac{1}{3x^2y} + \frac{2}{3x}$ w.r.t. x keeping y as constant we get

$$-\frac{1}{3xy} + \frac{2}{3} \log x.$$

Also in $\frac{1}{3xy^2} - \frac{1}{3y}$ term free from x is $-\frac{1}{3y}$ whose integral w.r.t. y is $-\frac{1}{3} \log y$. Hence the solution is

$$-\frac{1}{3xy} + \frac{2}{3} \log x - \frac{1}{3} \log y = \log C_1$$

$$\text{or } \log \frac{x^2}{y} = 3 \log C_1 + \frac{1}{xy}; \therefore \frac{x^2}{y} = C e^{1/xy}.$$

Ex. 64. Solve $y(x^2y^2 + 2) dx + x(2 - 2x^2y^2) dy = 0$.

Solution. Equation is of the form

$$yf(xy)dx + xg(xy)dy = 0.$$

$$\therefore \text{I.F.} = \frac{1}{Mx - Ny} = \frac{1}{3x^3y^3}$$

Multiplying by $\frac{1}{3x^3y^3}$, the equation becomes

$$\frac{x^2y^2 + 2}{3x^3y^2} dx + \left(\frac{2}{3x^3y^2} - \frac{2}{3y} \right) dy = 0, \text{ exact.}$$

$$\text{Now } \int \frac{x^2 y^2 + 2}{3x^3 y^3} dx = \int \left(\frac{1}{3x} + \frac{2}{3x^3 y^3} \right) dx = \frac{1}{3} \log x - \frac{1}{3x^2 y^3}$$

treating y as constant.

In coefficient of dy term free from x is $-\frac{2}{3y^3}$, whose integral w.r.t. y is $-\frac{2}{3} \log y$. Hence the solution is

$$\frac{1}{3} \log x - \frac{1}{3x^2 y^3} - \frac{2}{3} \log y = \log C, \text{ or } x = Cy^2 e^{1/x^2 y^3}.$$

Ex. 6. Solve $y(2xy+1)dx + x(1+2xy-x^2 y^3)dy=0$.

$$\text{Solution. I.F.} = \frac{1}{Mx - Ny} = \frac{1}{x^4 y^4}.$$

The equation after multiplying I.F. becomes

$$\left(\frac{2}{x^3 y^3} + \frac{1}{x^4 y^3} \right) dx + \left(\frac{1}{x^3 y^4} + \frac{2}{x^2 y} - \frac{1}{y} \right) dy = 0.$$

$$\text{Solution is } -\frac{1}{x^2 y^2} - \frac{1}{3x^3 y^3} - \log y = C.$$

Ex. 7 Solve

$$(xy \sin xy + \cos xy) y dx + (xy \sin xy - \cos xy) x dy = 0.$$

[Calcutta Hons. 59]

Solution. Equation is of the form

$$y f(xy) dx + x g(xy) dy = 0.$$

$$\therefore \text{I.F.} = \frac{1}{Mx + Ny} = \frac{1}{2xy \cos xy}.$$

Multiplying by I.F., the equation becomes

$$\frac{1}{2} \left(y \tan xy + \frac{1}{x} \right) dx + \frac{1}{2} \left(x \tan xy - \frac{1}{y} \right) dy = 0 \text{ (exact now).}$$

Integrating $\frac{1}{2} \left(y \tan xy + \frac{1}{x} \right)$ with respect to x , treating y as constant, we get $\frac{1}{2} [-\log \cos xy + \log x]$.

In coefficients of dy the term free from x is $-\frac{1}{2} \frac{1}{y}$ whose integral is $-\frac{1}{2} \log y$.

Hence the solution is

$$\frac{1}{2} [-\log \cos xy + \log x - \log y] = \text{constant}$$

or $x = cy \csc(xy)$.

Ex. 8. Solve $(y^4 - 2x^3 y) dx + (x^4 - 2xy^3) dy = 0$.

[Bombay 58]

Solution. The equation is homogeneous.

$$\text{I.F.} = \frac{1}{Mx + Ny} = \frac{1}{-xy(x^2 + y^2)}.$$

Now proceed yourself.